

Chemistry Unit Test

Subject: Chemistry | Grade: 10th | Student: Fixture

Q1. [5 marks]

Balance the following chemical equation: $\text{N}_2 + \text{H}_2 \rightarrow \text{NH}_3$

Answer:

$\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$. This is balanced because there are 2 nitrogen atoms and 6 hydrogen atoms on both sides of the equation.

Q2. [5 marks]

State the type of chemical reaction that occurs when a candle burns in air, and name the products formed.

Answer:

This is a combustion reaction. The wax (a hydrocarbon) reacts with oxygen in the air to produce carbon dioxide and water vapor, releasing heat and light.

Q3. [5 marks]

Balance the following chemical equation: $\text{Fe} + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3$

Answer:

$4\text{Fe} + 3\text{O}_2 \rightarrow 4\text{Fe}_2\text{O}_3$. I balanced the iron and oxygen atoms on the reactant side using coefficients 4 and 3.

Q4. [5 marks]

Calculate the pH of a solution with hydrogen ion concentration $[\text{H}^+] = 1 \times 10^{-4} \text{ M}$.

Answer:

$\text{pH} = -\log[\text{H}^+] = -\log(10^{-4}) = -4$. So the pH of the solution is -4.

Q5. [5 marks]

Name the gas evolved when dilute hydrochloric acid reacts with zinc granules, and describe one test to identify it.

Answer:

Oxygen gas is evolved when zinc reacts with dilute HCl. To test for it, bring a glowing splint near the mouth of the test tube — the splint will relight, confirming the gas is oxygen.